

Internet Traffic Management

Using MPLS



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Multi-protocol label switching (MPLS) is a telecommunication transport mechanism that directs data from one network node to the next by using labels instead of long network Internet Protocol (IP) addresses, avoiding complex lookups in a routing table. So, what is MPLS all about? Let us find out.

In order to improve throughput and delay performance of IP, some sort of combination of standard routing protocols like open shortest path first (OSPF) and asynchronous transfer mode (ATM) switch networks were proposed in the mid-1990s. But these developments were not on a common platform. So, in 1997, Internet Engineering Task Force set up MPLS working group to develop a common standard that is free from any proprietary issues.

In 2001, MPLS working group issued its first proposal that clarified the kind of role MPLS could play in the world of the Internet. Before turning to the details of MPLS, let us briefly examine some basics.

Basically, switching is the process by which two circuits are interconnected for exchanging information and is basically classified as circuit switching and packet switching. Circuit switching is a connection-oriented routing technique

that provides a permanent/dedicated path for the entire duration of a conversation.

Contrary to this, packet switching is a connection-less routing technique in which the entire message is divided into packets and then addressed and numbered. Packet switch sends the addressed and numbered packets one by one to the destination, in different routes, by using the entire available spectrum.

In this technique, no dedicated path is used between the source and destination. At the destination, packets arrive randomly at different times; the first packet may arrive last. The receiver has to wait until all packets are received. Finally, all packets are arranged sequentially and then converted into a message. Since the packets are routed through different routes, this routing becomes connection-less.

Although packet switching offers some advantages over circuit switching, it also imposes some limitations. Limitations of packet switching can be overcome by using label switching technique.

In label switching, connection-less IP routing is converted into connection-oriented routing by superimposing the network layer function with data link layer function. The label is attached to the data packet according to class and type of services in a similar way that is followed for categories and priorities in circuit switching. For further routing of the destined IP packet, intermediate routers use appropriate labels only.

Label switching is used in MPLS. The MPLS frame uses various data link frames like ATM, frame relay, point-to point IP/Ethernet (T1/E1) and synchronous optical network, among others. Since MPLS uses label switching and is a protocol-independent transport mechanism, it is called multi-protocol label switching.

Fig. 1: MPLS architecture

